MOHAMAD KHALIF

Geodetic Engineer | GIS & Remote Sensing Analyst | Geospatial Data Analyst
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PROFESSIONAL SUMMARY

Geodetic Engineering graduate from Institut Teknologi Nasional Malang with hands-on experience in topographic surveying, construction stake-out, and cut-and-fill volume calculation on the Probolinggo-Banyuwangi Toll Road project, plus six months of inventory and operational data analysis for a 15-service-center after-sales network. Proficient in GIS and remote sensing (ArcGIS, QGIS, Google Earth Engine, ENVI), UAV photogrammetry (Agisoft Metashape), AutoCAD Civil 3D, PostgreSQL/PostGIS, and Python for spatial analysis, automation, and machine learning. Detail-oriented and data-driven, with a record of turning field and operational data into accurate maps, reports, and KPI insights.

CORE SKILLS

- **GIS & Remote Sensing:** ArcGIS (ArcMap), QGIS, Google Earth Engine, ENVI, Global Mapper, Spatial Analysis, Image Classification, NDVI, Multitemporal Analysis, Map Layout
- **Surveying & CAD:** Topographic Survey, Stake Out, Cut and Fill Volume Calculation, Long & Cross Sections, AutoCAD Civil 3D, GNSS/GPS Geodetic, Total Station, Automatic Level (Waterpass), Theodolite
- **Photogrammetry:** Agisoft Metashape, UAV Aerial Mapping, GCP/ICP, Orthophoto, DEM/DSM, 3D Modeling
- **Programming & Databases:** Python (pandas, NumPy, rasterio, scikit-learn, XGBoost, TensorFlow/Keras), JavaScript, MATLAB, SQL, PostgreSQL/PostGIS, Jupyter Notebook
- **WebGIS:** GeoServer (WMS/WFS), Leaflet, OpenStreetMap
- **Data Analysis & Reporting:** Microsoft Excel (Pivot Table, VLOOKUP), Power BI, Microsoft PowerPoint, Microsoft Word, KPI Reporting, Forecasting, Inventory Management
- **Soft Skills:** Attention to Detail, Analytical Thinking, Cross-team Coordination, Time Management, Adaptability

PROFESSIONAL EXPERIENCE

Content Moderator **Jun 2026 - Sep 2026**
TELUS Digital (formerly TELUS International) | Denpasar, Bali

- Reviewed and moderated user-generated content (text, images, and video) against client community and policy guidelines in a 24/7, shift-based operation.
- Applied detailed decision criteria consistently to meet individual accuracy, quality, and productivity targets.
- Flagged and escalated policy violations and edge cases, documenting decisions for quality review and reporting.
- Handled sensitive and confidential data in strict compliance with NDA, data privacy, and standard operating procedures.

Warehouse Admin & Data Analyst **Dec 2025 - May 2026**
PT Thousand Island Services (realme Indonesia After-Sales Service Agent) | Denpasar, Bali

- Managed end-to-end spare-part inventory (receiving, verification, storage, allocation, and distribution) for 15 realme service centers across Bali, Nusa Tenggara, Kalimantan, Sulawesi, Maluku, and Papua.
- Contributed to raising the area's 24-hour repair closing rate from 84% to 92% (Dec 2025 - Feb 2026) by analyzing pending work orders and prioritizing spare-part allocation for high-demand models.
- Sustained a Key Availability score of 100 (Jan - Feb 2026) through daily stock monitoring, consumption-trend analysis, import planning, and inter-agent / SC-to-SC stock transfers.
- Automated recurring reports with Python (pandas, NumPy) for Key Availability, inventory turnover (ITO), and pending work orders, and prototyped XGBoost and hierarchical models to forecast monthly spare-part consumption.
- Coordinated monthly stock opname for spare parts, devices, IoT products, and assets; reconciled physical counts against system records and issued final variance reports.
- Dispatched all warehouse-to-service-center shipments within 24 hours and kept defective-material returns (RTC) within the 35-day limit through a bi-weekly return schedule.
- Processed DOA/DAP warranty claims, BMT factory returns, special exchanges, and customer refunds, ensuring complete documentation to minimize financial loss.
- Prepared weekly and bi-weekly KPI presentations and monthly GMI reports (Key Availability, RO completion, ITO, closing rate, 6S) for management.

Assistant Surveyor Intern **Jul 2024 - Sep 2024**

PT Virama Karya (Persero) | Probolinggo-Banyuwangi Toll Road Package 1, East Java

- Staked out Top Subgrade, Lean Concrete, and Rigid Pavement layers to ensure precise alignment with design coordinates and elevations.
- Conducted topographic surveys and processed field measurements into surfaces, long sections, and cross sections using AutoCAD Civil 3D.
- Calculated cut-and-fill earthwork volumes for side-drainage design at STA 5+000 - 6+000 and documented the method in the internship report "Terrestrial Survey Strategy for Determining Cut and Fill Volumes of Side Drainage."
- Supported daily survey activities and field data documentation for toll road construction supervision.

SELECTED PROJECTS

- **Cut and Fill Volume Analysis - Land Development, Lombok (2025):** Processed field survey points into a TIN surface and contour map using AutoCAD Civil 3D and Global Mapper; computed 1,135 m³ cut and 8,496 m³ fill volumes against an 8 m MSL design elevation.
- **Undergraduate Thesis - PM2.5 Mapping, Jakarta Metropolitan Area (2021-2024 data):** Analyzed PM2.5 main-component concentration from multitemporal Sentinel-2A and Landsat-8 OLI imagery using Dispersion Coefficient, Angstrom Exponent, and NGAI in MATLAB; validated with AERONET AOD data (BMKG Jakarta).
- **PTSL Land Registration Survey - Solokuro, Lamongan (2023):** Mapped the village boundaries of Bluri and Dagan with local residents and supervised the installation of Independent Control Points (ICP) for UAV aerial photography.
- **UAV Aerial Mapping & Orthophoto:** Generated dense point cloud, DEM, and orthomosaic in Agisoft Metashape with GCP/ICP control, achieving 2.09 cm/pixel ground resolution.
- **WebGIS Development:** Built an interactive web map using PostgreSQL/PostGIS as the spatial database, GeoServer (WMS/WFS) for map services, and Leaflet/OpenStreetMap for visualization.
- **Mangrove Change Detection - Pemogan, Bali (2014-2023):** Processed Landsat and Sentinel imagery in Google Earth Engine (cloud masking, NDVI) and classified mangrove density and area change in ArcMap.
- **Deep Learning Vegetation Classification:** Developed a CNN model with TensorFlow/Keras to classify and map vegetation from high-resolution imagery using Python (rasterio, NumPy, scikit-image).

EDUCATION

Bachelor of Engineering (B.Eng.) in Geodetic Engineering **2021 - 2025**

Institut Teknologi Nasional Malang (ITN Malang) | Malang, East Java | GPA 3.15/4.00

CERTIFICATIONS & TRAINING

- **Level Up Your Spatial Analysis: From QGIS to Python** - MAPID Academy (Oct 2026)
- **Internship Completion Certificate** - PT Virama Karya, Probolinggo-Banyuwangi Toll Road Package 1 (2024)
- **TOEFL Prediction Test** - Language Laboratory Unit, ITN Malang, Score 460 (Aug 2025)

ORGANIZATIONAL EXPERIENCE

- **Geodetic Engineering Student Association (Himpunan Mahasiswa Geodesi), ITN Malang:** Publication, Decoration & Documentation (PDD) Coordinator for the IMGI Anniversary; PDD Member for the 38th Geodetic Engineering Anniversary (2023); committee member for Geodesi Menyapa and guest lecture events.

LANGUAGES

Indonesian (Native) | English (Intermediate)